

Project Proposal

Overview

Customer churn refers to the phenomenon that customers stop using a company's products or services during a certain period of time. While churn is a natural part of any business cycle, high customer churn rates can be a significant problem, indicating poor customer retention and low customer loyalty. A high churn rate often relates to a lot of potential issues, such as poor product or service quality, bad product-market fit, price disadvantage, limited customer support, intense competition, etc. Customer churn is not only loss of customers, but also affects the long-term profitability and sustainability of a company.

Customer churn is a particularly pressing issue in telecommunications companies, which are service providers in the networking industry that facilitate the electronic transmission of information through telephony and data communication services. With the advent of mobile telephony, most telcos also provide internet service. Telco customer churn rates are among the highest in any industry, averaging 31% in 2024 (Tessitore). This is due to the inherent nature of this sector, such as subscription-based plan, highly competitive market of providers offering similar products, rapid technological advancement, etc. Customer churn can lead to loss of revenues and even a business crumble. Thus, understanding and predicting customer churn is very important, because it allows companies to proactively address at-risk customers with targeted strategy.

This project focuses on utilizing the Telco Customer Churn dataset

(<https://huggingface.co/datasets/scikit-learn/churn-prediction>) from Hugging Face to explore churn analysis and prediction. This dataset includes the following information:

- Customers who left within the last month: the column is called Churn.
- Services that each customer has signed up for: phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies.
- Customer account information: how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges.
- Demographic info about customers: gender, age range, and if they have partners and dependents.

The primary objective of this project is to build predictive machine learning models that accurately classify customers into "churn" and "non-churn" categories. This project will also include exploratory data analysis to summarize main characteristics, analyze influential factors contributing to customer churn, and visualize the relationships between these factors and churn. The insights derived can help support data-driven strategies to reduce customer churn rates.

Prior Work

Machine learning approaches have become essential tools in predicting customer churn. Supervised machine learning techniques form the foundation of most churn prediction systems. Logistic regression, widely recognized for its computational simplicity and model interpretability, remains a popular choice for binary classification problems (Burez and Van den Poel). Decision trees and their ensemble variations, such as Random Forests, provide robust performance by capturing non-linear patterns and aggregating results across multiple classifiers (Lalwani et al.).

In recent years, ensemble learning techniques have gained prominence. Boosting methods, such as AdaBoost and XGBoost, have been particularly effective in addressing class imbalance—a common issue in churn datasets. Studies highlight the superior performance of XGBoost, consistently outperforming traditional methods in terms of predictive performance. Ensemble techniques not only improve prediction outcomes but also mitigate overfitting by aggregating predictions from multiple models (Lazarov and Capota; Lalwani et al.). Thus, logistic regression, decision tree, random forest, XGBoost and AdaBoost are potential methods that can be used in our project.

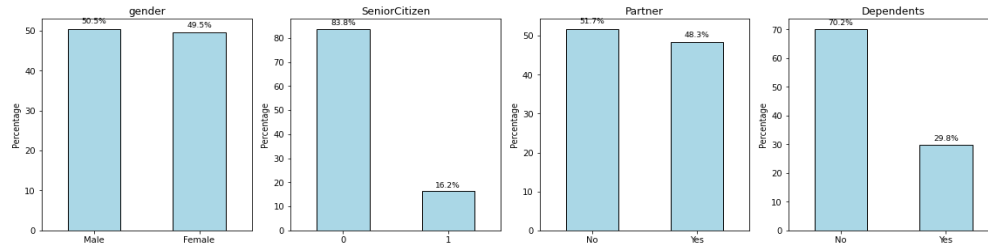
Evaluation metrics are also essential in churn prediction. Metrics such as ROC and AUC are particularly valuable for their threshold-independent evaluation capabilities and their robustness to imbalanced datasets, offering a more comprehensive assessment than traditional accuracy measures (Lalwani et al.; Altairey et al.).

Preliminary Results

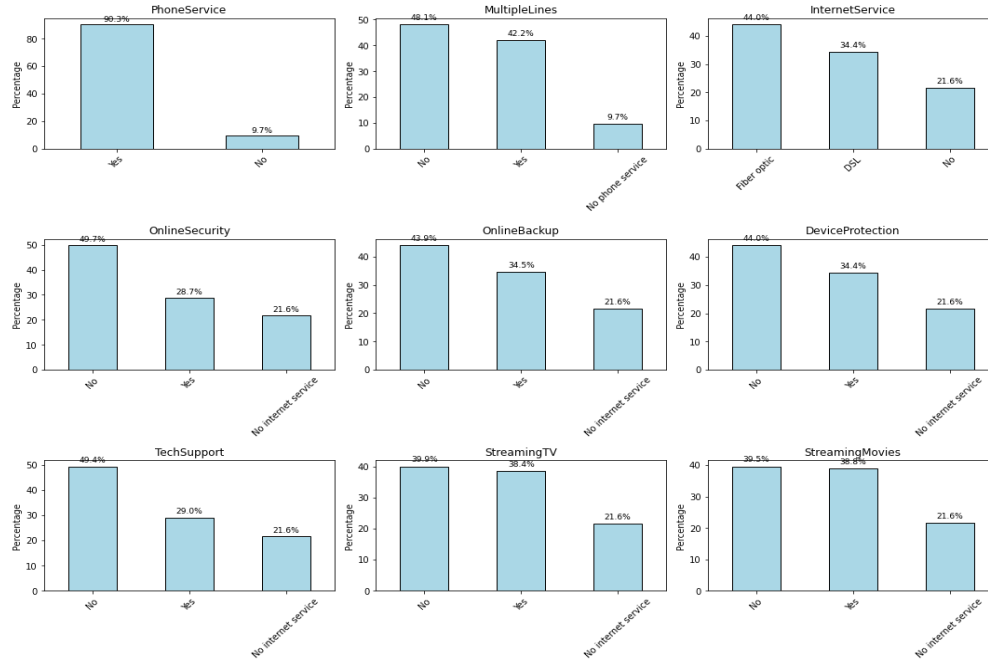
This dataset contains both numerical and categorical variables. The dataset quality is generally good, with minimal preprocessing required. The variable TotalCharges was incorrectly identified as an object type due to non-numeric entries. After converting it to a numerical variable, we found 12 missing values, all corresponding to customers with tenure = 0. Since tenure = 0 only occurs when TotalCharges is NA in the whole dataset, this pattern indicates that the missing TotalCharges values are associated with customers who just started their service in less than a month, so there are no total charges yet. These missing values can be replaced with 0 or removed without impacting the analysis. Other variables show no missing values or data quality issues.

To gain an understanding of the data and get initial insights, univariate and bivariate analysis were performed. For univariate analysis, bar plots were used to explore the distributions of categorical variables, while histograms were utilized to assess the distributions of numerical variables. For bivariate analysis, to examine the relation between variables and churn, bar plots were employed to provide a comparison of churn rates across different categories, and boxplots were used to show differences in the distributions of numerical features for churned and non-churned customers.

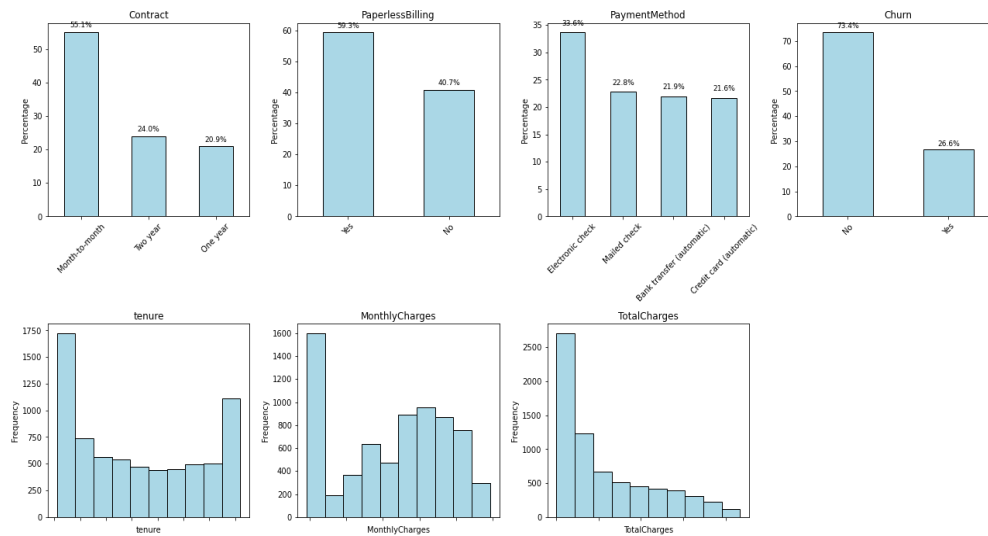
Demographic Features



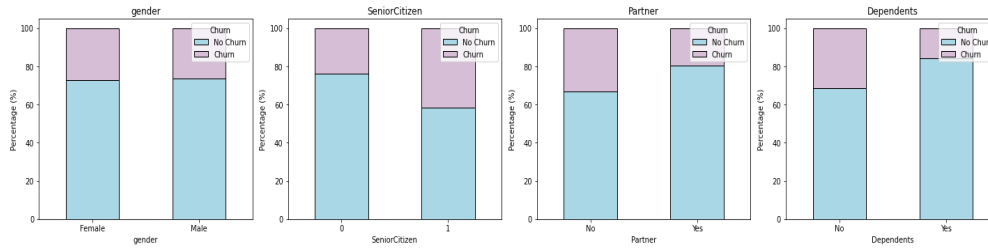
Service-Related Features



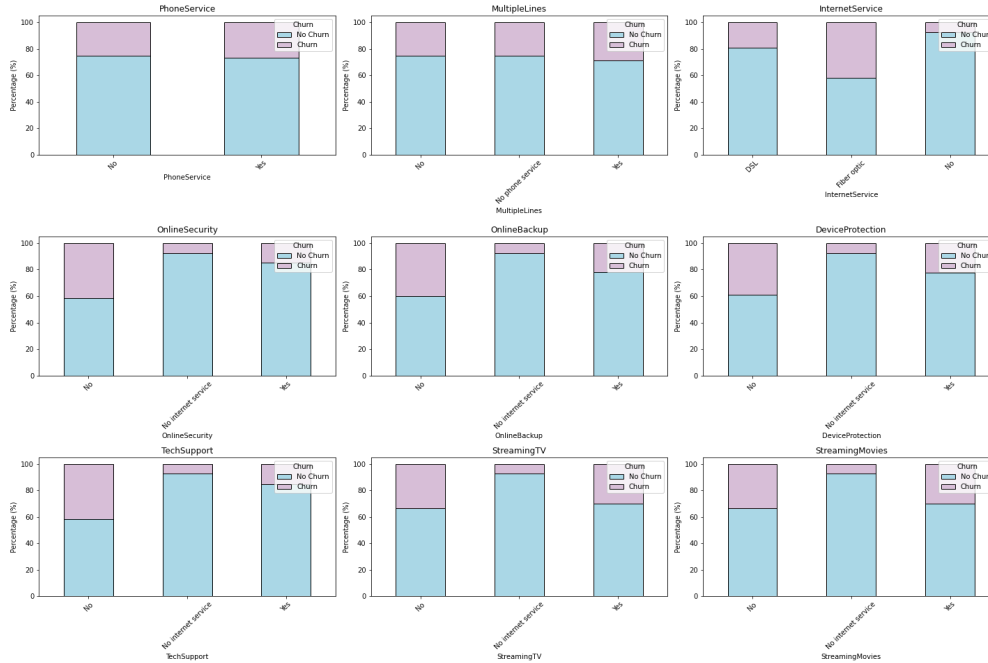
Account-Related Features



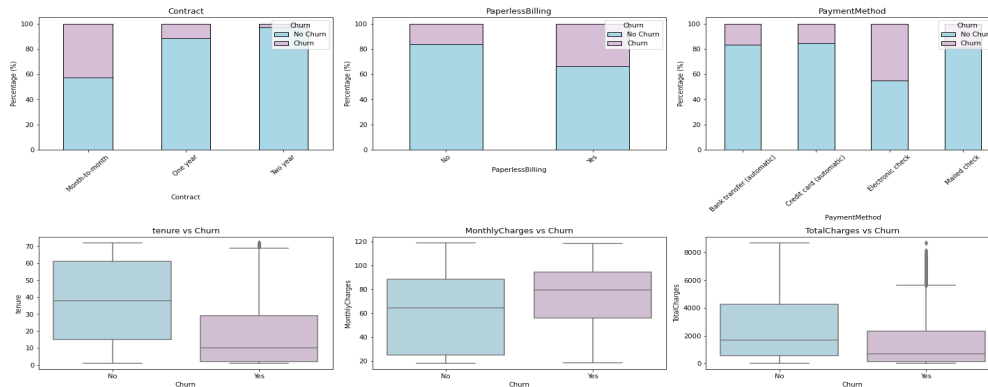
Demographic Variables with Customer Churn



Service Variables with Customer Churn



Account Variables with Customer Churn



Bivariate analysis revealed pattern and relation between explanatory variables and churn. For demographic factors, we have the following observations: churn rates are similar across genders, but senior customers, single customers, and customers without dependents are more likely to churn. For service factors, we have the following observations: churn rates are similar across customers with and without phone services and across customers with and without

multiple lines and without phone services; fiber optic internet users exhibit significantly higher churn compared to DSL or no internet service users; customers who do not subscribe to value-added services (Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies) tend to have higher churn rates than those with these services, those with these services tend to have higher churn rates than those without internet services at all. For account variables, we have the following observations: month-to-month contract customers have the highest churn rates, followed by one-year contracts, and customers with two-year contracts churn the least; customers paying via electronic checks show the highest churn than bank transfer (automatic), credit cards (automatic), and mailed checks; paperless billing, shorter tenure, and higher monthly charges are associated with higher churn.

Additionally, there are potential challenges we need to deal with. The dataset has significantly fewer churn cases compared to non-churn cases, making churn prediction challenging. Imbalanced classes could lead models to consistently predict the majority class and have poor performance on the minority class. Secondly, numerical variables like tenure, MonthlyCharges and TotalCharges are not normally distributed and may require normalization for better model performance.

For the baseline model, logistic regression was chosen as it is a simple and straightforward algorithm that can produce interpretable results. It is also appropriate for binary classification problems. However, logistic regression has limitations when applied to imbalanced datasets, as it tends to predict the majority class more frequently. We can explore other machine learning models to better deal with this problem. Here is the result from logistic regression. Precision measures the proportion of correctly predicted positive observations out of all observations predicted as positive. For class 0 (Non-churn): Precision is 0.84, indicating 84% of the predictions for class 0 are correct. For class 1 (Churn): Precision is 0.62, indicating 62% of the predictions for class 1 are correct. Recall measures the proportion of correctly predicted positive observations out of all actual positives. For class 0 (Non-churn): Recall is 0.89, meaning 89% of the actual instances of class 0 are correctly predicted. For class 1 (Churn): Recall is 0.53, which is relatively low, indicating that only 53% of the actual instances of class 1 are correctly predicted. This reflects the difficulty in predicting the minority class. The F1-score combines precision and recall and is useful when the dataset is imbalanced. For class 0 (Non-churn): The F1-score is 0.87, reflecting a good balance between precision and recall for the majority class. For class 1 (Churn): The F1-score is 0.58, indicating weak performance in predicting the minority class due to low recall and precision.

	precision	recall	f1-score	support
0	0.84	0.89	0.87	1041
1	0.62	0.53	0.58	366
accuracy			0.80	1407
macro avg	0.73	0.71	0.72	1407
weighted avg	0.79	0.80	0.79	1407

For tools from class, Matplotlib and Seaborn will be used for exploratory data analysis (EDA) and visualizing relationships between variables and churn, and Pandas will be used for structured data manipulation. I will explore tools such as Scikit-learn for supervised and unsupervised machine learning techniques and evaluation metrics.

Project Deliverables

A successful project will produce high-performing churn prediction models which can accurately classify customers as “churn” or “non-churn,” and robust diagnostics to evaluate its effectiveness. The model’s performance will be assessed using various metrics, with particular emphasis on the F1 score, which balances precision and recall. We will prioritize recall to ensure the model can effectively identify customers at risk of churning, minimizing false negatives where churners might be overlooked. Besides predictive accuracy, the project seeks to develop a model that is interpretable, allows for the identification of influential features, such as contract type, payment method, or tenure, that contribute to churn behavior. By highlighting these factors, the project will provide insights to help inform data-driven decisions and support customer churn management. Last but not least, a successful project should effectively convey the results in a clear and understandable manner, ensuring that the insights are accessible to both technical and non-technical stakeholders. To achieve this, the project will utilize compelling visualizations, such as bar plots, heatmaps, and boxplots, to highlight patterns and relationships in the data, making complex findings more intuitive.

Works Cited

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